Baton

An evidence-linked, permission-aware knowledge handover platform

Round	Round 2 - The Build
Team	ByteStorm
Participant	Ritika Bhati
Event	CodeFrontier Hackathon | We Code Coders
Date	8 November 2026
Live demo	https://baton-t3ax.onrender.com
Source code	https://github.com/Ritika-Bhati-2266/Baton

This report covers the functional product, implementation details, changes from the Round 1 plan and the three committed Round 3 improvements. The product is a working end-to-end MVP; the limitations section is deliberately explicit.

1. Submission summary

Baton turns scattered, authorised project activity into a reviewed continuation plan for one departing member and one incoming member. The unit of trust is a single claim with a checked quote, a visible evidence label and an access label. A human resolves conflicts, answers or accepts gaps, and approves the handover before anything is shared.

Area	What is working	How a judge can see it
Core workflow	Sources to reviewable, evidence-linked handover without a broken step	Run the seeded rover-club case end to end
Trust model	Deterministic quote verification, evidence labels, conflict and gap detection	The review console shows VERIFIED, SINGLE SOURCE, CONFLICTING and MISSING claims
Human control	Per-claim approve, reject, edit, conflict resolution, gap answer/accept	Every action is available in the review console and written to the audit log
Permissions	Source ACLs, claim inheritance, server-side role filtering	Compare the incoming, departing and approver handover views
Publishing	Unresolved conflicts and critical gaps block sharing	The publish button is disabled with explicit reasons until decisions are made
Reliability	Automated end-to-end smoke test	scripts/smoke_test.py prints SMOKE OK

Links

- **Working project:** https://baton-t3ax.onrender.com
- **Source code:** https://github.com/Ritika-Bhati-2266/Baton
- **Health check:** /health
- **API state:** /api/cases/rover-demo/state
- **OpenAPI docs:** /docs

Judge's 90-second path

- Open the dashboard and click **Open the demo review console**.
- Inspect the planted motor-driver conflict, the two critical gaps and the labelled source inventory.
- Click **Keep this side** on the driver-A side, then answer or accept the critical gaps.
- Click **Approve all reviewable**, then **Publish handover**.
- The incoming person's view opens. Restricted claims are counted, not rendered; each visible claim links to its quote.
- Switch the role tabs to compare the incoming, departing and approver views, then open the Audit page.

2. Problem and users

When a person leaves a project, the files usually stay. What leaves is the ability to interpret them. A repository shows what the code does but not why a design was chosen. A task board shows tickets but not which ones are quietly blocked. A chat history holds a promise to a vendor, buried in a long thread.

Knowledge that leaves	Where it hides	What goes wrong after the exit
Decisions and reasoning	Chat threads, meeting notes, PR comments	The successor re-opens settled debates or reverses a decision for the wrong reason

Knowledge that leaves	Where it hides	What goes wrong after the exit
Unfinished work	Half-merged branches, stale tickets, personal to-do lists	Work is duplicated or silently dropped
Dependencies and commitments	Emails, calls, verbal promises	Deadlines and obligations are missed
Recurring problems and workarounds	The leaver's memory	The same failure is rediscovered expensively
Who to ask	Informal relationships	The successor does not know which contact or reviewer to approach

Users

User	Role in a handover	Pain today	What Baton gives them
Incoming person	Takes over ownership	Manual digging; fear of missing a hidden commitment	A continuation plan with owned items, risks, open questions and first-week actions, each with sources
Departing member	Knows the context	No time; documents from memory; privacy worry	An auto-built draft to correct, plus targeted gap questions
Team lead / approver	Accountable for continuity	Cannot verify a handover or see what is missing	A review console with evidence, conflicts, gaps and per-claim decisions
Admin / security owner	Guards access	Fear that tools leak confidential chats or credentials	Source ACLs, redaction preview, audit log and a publish gate

Scope. Baton solves the handover of one project from one departing member to one incoming member, with the consent of the people who own the sources. It deliberately does not solve company-wide knowledge search, HR offboarding or live employee monitoring.

3. The core idea: claims, not paragraphs

A fluent summary that invents an owner or hides a conflict looks authoritative and is more dangerous than no summary. Baton therefore stores a **Claim Ledger**: a list of typed claims (decision, owner, risk, dependency, open question, next action, blocker, recurring problem, contact). Every claim keeps the exact supporting quote, the source ID, the date and author, an evidence label and an access label.

Evidence labels

Label	Meaning	How Baton decides
VERIFIED	Quote exists and is backed well enough	Deterministic substring check passes, and the claim rests on a primary source or two independent informal sources
SINGLE SOURCE	Plausible but rests on one informal mention	Quote check passes, but the only support is one notes or chat source
CONFLICTING	Two sources disagree	Same topic key, different value; both sides stay visible
MISSING	Expected knowledge has no evidence	Gap rule, e.g. a task with no owner or a dependency with no contact
HUMAN ASSERTED	Added or changed by a person	Edits and gap answers; never shown as machine-verified
INFERRED	Conclusion drawn from checked quotes	Supporting quotes verified in code; reviewer confirmation required
QUOTE FAILED	Quote not found in the cited source	Kept out of the reader view; appears in the reviewer queue

Trust gates

- **Permissions are inherited from sources.** A claim takes the most restrictive access label of its evidence.
- **No machine claim without a checked quote.** The verifier is deterministic code, not a model's opinion.
- **Gaps are first-class output.** Baton reports what it could not find instead of inventing an answer.
- **Nothing is shared until a human approves it.** The incoming person sees approved claims only.

4. Complete workflow

Step	What the user does	What the system does	Output
1. Connect	Creates a handover case, picks sources and sets access labels	Read-only ingestion, stable chunking, redaction preview	Indexed, labelled source chunks
2. Reconstruct	Waits for the pipeline	Deterministic extraction of typed claims with quote and source	Draft Claim Ledger
3. Verify	Reviews the label dashboard	Quote match, conflict detection by topic key, gap rules	Claims with evidence labels and a gap list
4. Review	Edits, rejects, approves, resolves conflicts, answers or accepts gaps	Stores reviewer identity and time; blocks publishing while blocking issues remain	Approved, versioned ledger
5. Handover	Opens a role-appropriate view and acknowledges receipt	Filters claims by the reader's permissions; logs access and acknowledgement	Continuation plan plus receipt

Demo scenario: the rover club

A student club is handing over its autonomous-rover project from A. Rao to N. Singh. The synthetic dataset contains a README, meeting notes, a commit export, a task CSV, a leads-only chat export and a private budget note. The following failures are planted so they can be tested repeatably:

- **Conflict:** the 12 October meeting decided to move to motor driver B, but the README and the latest commit still configure driver A.
- **Critical gap:** the spare-driver task has no owner.
- **Critical gap:** the demo-venue dependency has no contact.
- **Non-critical gap:** no rollback procedure is documented.
- **Permission test:** the vendor chat is leads-only and the budget note is private.

5. Implemented features

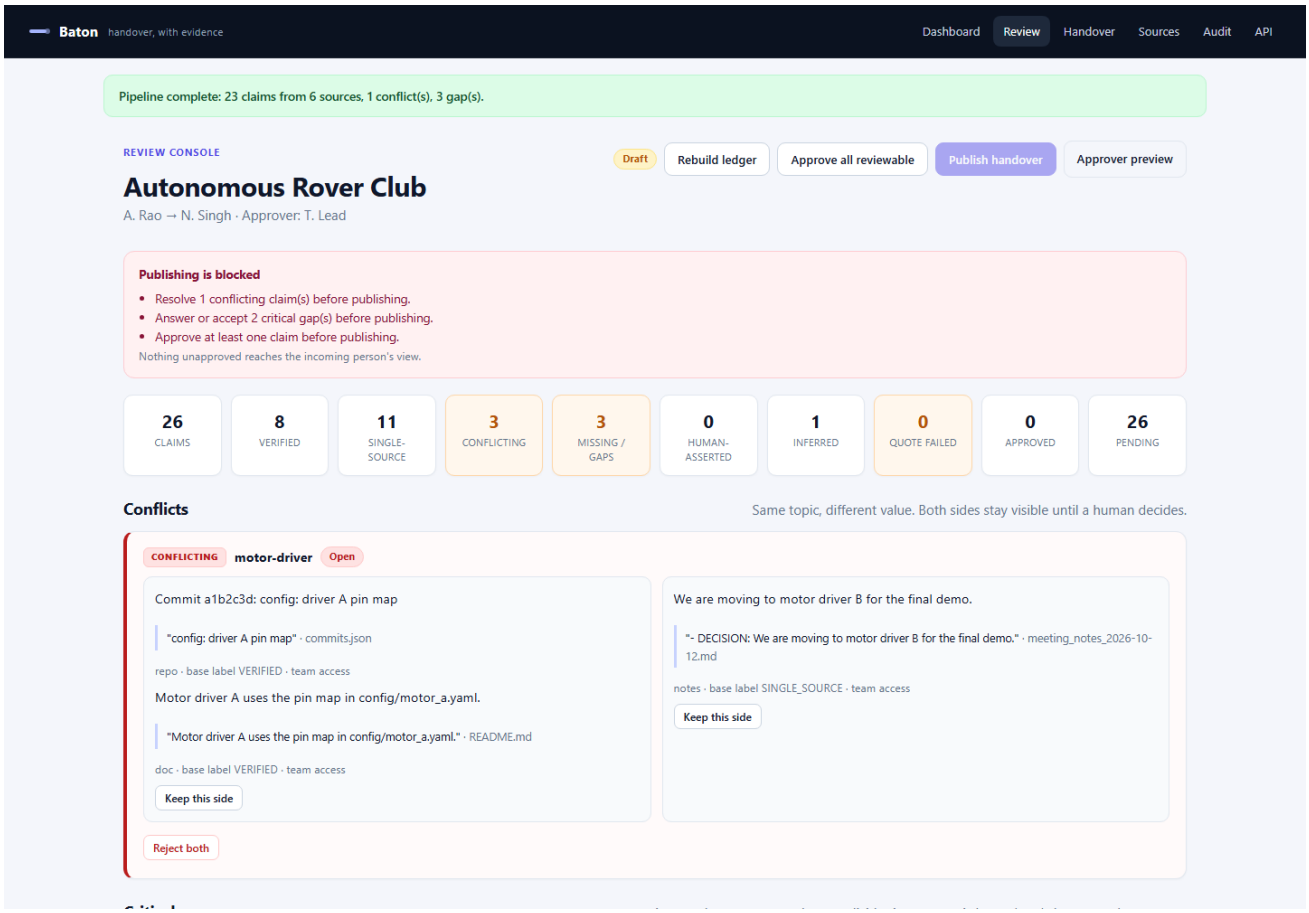
Round 2 Must features

Feature	Status	Where it lives
Document, Markdown, text, CSV, JSON and PDF upload	Done	Sources page; /cases/{id}/sources/upload
Read-only GitHub import of public issues and commits	Done	POST /cases/{id}/sources/github
Typed claim extraction with quote, source, date and author	Done	app/extract.py, app/pipeline.py
Deterministic quote verifier	Done	app/verify.py
Evidence labels, conflict detection and gap rules	Done	app/verify.py
Review console: edit, reject and approve per claim	Done	review.html, /claims/{id}/... routes
Conflict resolution: keep a side or reject both	Done	POST /conflicts/{id}/resolve
Role-based views with source ACLs	Done	app/verify.py, handover.html
Secret stripping and identity pseudonymisation preview	Done	app/redaction.py, Sources page
Append-only audit log	Done	audit table, Audit page
Publishing gate for unresolved conflicts and critical gaps	Done	publish_rules, Publish button

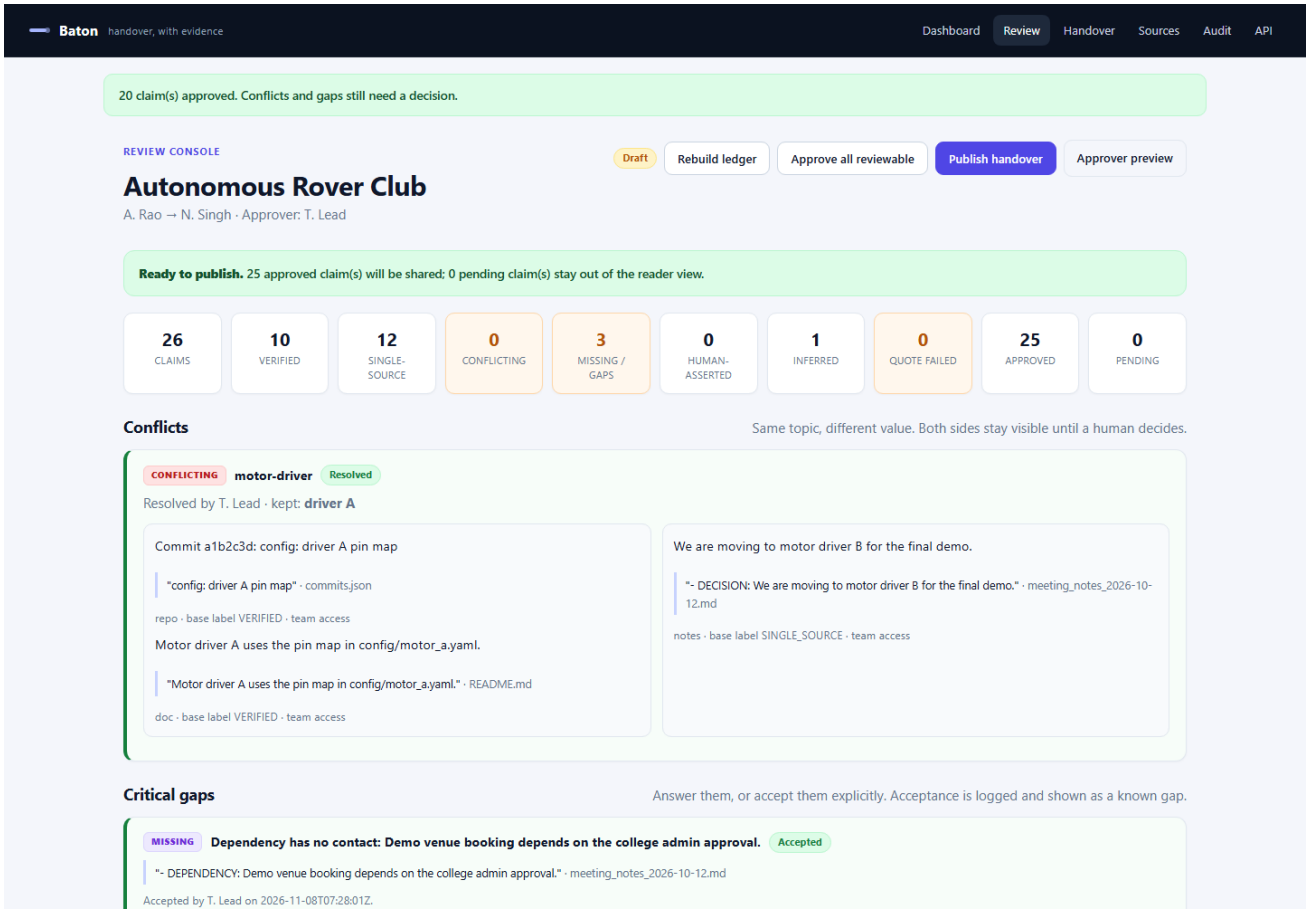
Additional features delivered

- Acknowledgement of receipt, with the acknowledgement stored on the case and in the audit log.
- Human answer to a gap, stored as HUMAN ASSERTED evidence.
- Inferred claims that require explicit reviewer confirmation.
- JSON state API and automatic OpenAPI documentation.
- Docker and Render deployment configuration.
- Automated end-to-end smoke test.

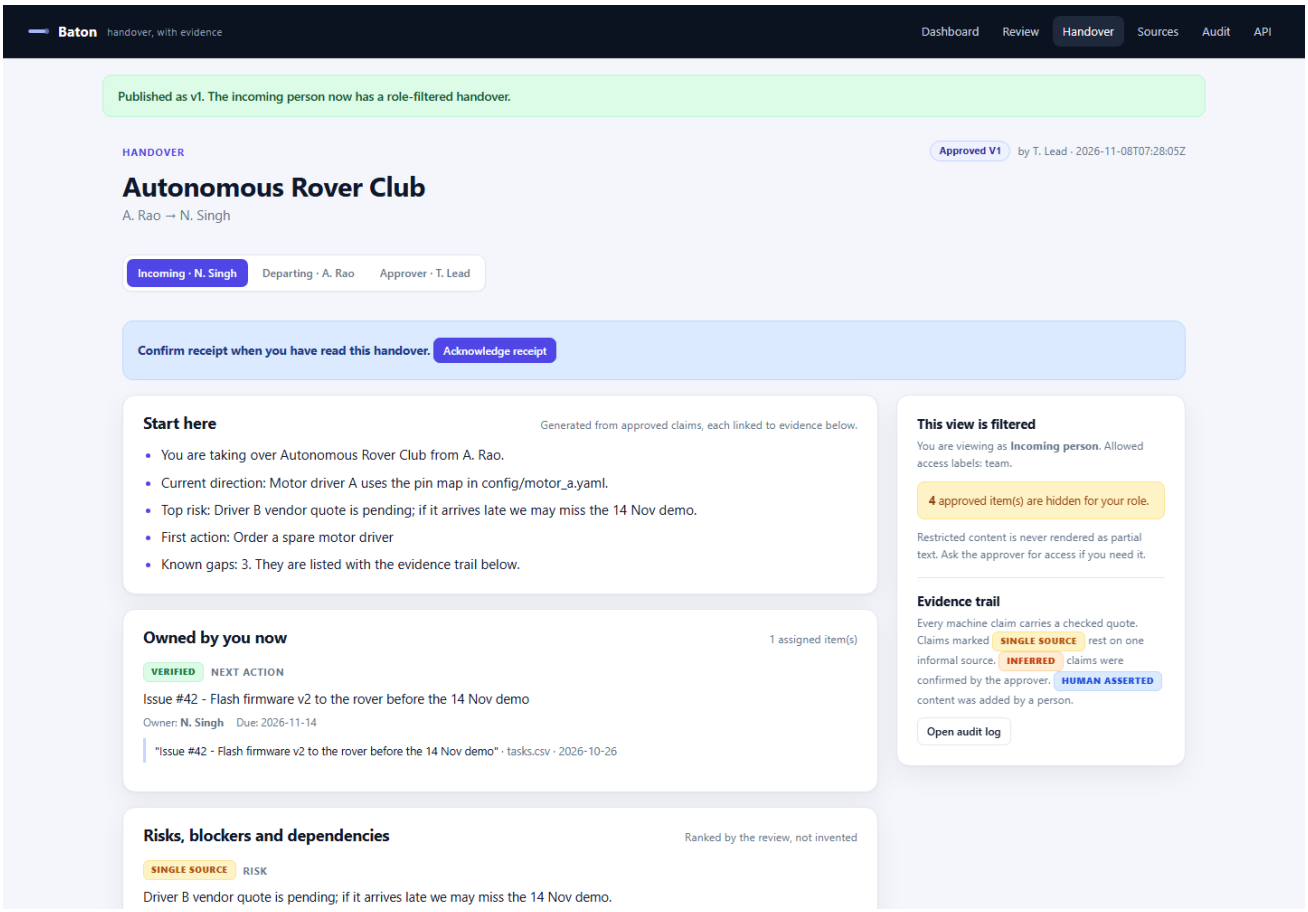
Screens in the working product



Review console: blocking issues, evidence labels, conflicts and critical gaps.



After the human decisions: conflict resolved, gaps handled, claims approved.



Incoming-person view: filtered, evidence-linked and approved.

6. Architecture and implementation

```
Browser
  |  server-rendered HTML + small JS
  v
FastAPI service
  |-- ingestion: upload / GitHub / demo seed
  |-- chunking + redaction preview
  |-- deterministic extraction (typed claims + quotes)
  |-- deterministic verifier (quote match, conflicts, gaps)
  |-- SQLite claim ledger (claims, evidence, conflicts, audit)
  |-- review actions + publish gate
  +-- role-filtered handover views
```

Layer	Choice	Why
Backend	FastAPI + Python 3.12	Typed request/response handling, automatic OpenAPI docs, strong text-processing ecosystem
Templates	Jinja2	Server-rendered review and reader views with no separate build step
Storage	SQLite (WAL)	One deployable service for the demo; transparent ledger for judges
Parsing	pypdf, csv, json	PDF, task CSV and repository export ingestion
GitHub	httpx + GitHub REST API	Read-only public import; optional GITHUB_TOKEN for private repos
Deployment	Docker + Render	One command local setup and a public judge-accessible link
Tests	FastAPI TestClient	Full journey assertions without a browser

Data model

Table	Purpose
cases	Handover case, status, published version and acknowledgement
sources	Authorised inputs with kind, ACL, author and source date
chunks	Stable source windows with character offsets
claims	Typed claims with value, owner, due date, event date, basis, evidence label and status
evidence	Quote, chunk, source and verification result
conflicts	Topic-level disagreement and its resolution
audit	Append-only record of approvals, edits, resolutions, publishing and access events

Key algorithms

- **Stable chunking:** sources are split into overlapping windows with permanent IDs and character offsets.
- **Typed extraction:** explicit markers, README ownership sentences, run instructions, commit/issue JSON, task CSV rows and chat exports produce claims with exact quotes.
- **Deterministic verification:** whitespace-normalised substring check against the cited source; failure downgrades the claim to QUOTE FAILED.
- **Conflict detection:** claims on the same value topic with different values are paired and marked CONFLICTING; both sides remain visible.
- **Gap rules:** an unowned task topic, a dependency with no contact and a missing rollback procedure produce MISSING claims.
- **Access filtering:** claims inherit the most restrictive source ACL; role views filter in code, never by re-prompting a model.
- **Publish gate:** open conflicts, unaccepted critical gaps and zero approved claims block publishing.
- **Audit:** every review action is appended with actor, role, action, detail and timestamp.

API surface

Method and path	Purpose
GET /	Dashboard and case list
GET /cases/{id}/review	Review console with filters
POST /cases/{id}/build	Run ingestion, extraction, verification, conflicts and gaps
POST /claims/{id}/approve, /reject, /edit	Per-claim human decisions
POST /conflicts/{id}/resolve	Keep one side or reject both
POST /claims/{id}/accept-gap, /answer-gap	Handle missing knowledge explicitly
POST /cases/{id}/publish	Enforce the publish gate and increment the version
GET /cases/{id}/handover?role=...	Role-filtered continuation plan
GET /cases/{id}/audit	Append-only audit trail
GET /api/cases/{id}/state	Machine-readable ledger state
GET /health	Deployment health check

7. Permissions, privacy and security

Role	Allowed access labels	What they see
Incoming person	team	Approved team-labelled claims only; restricted items become a count, never partial text
Departing member	team, leads	Team and leads-only claims; private finance notes remain hidden
Approver	team, leads, private	The full ledger, including private evidence, to make an informed decision
Admin	team, leads, private	Full ledger plus audit access

- **Connecting a source does not make it shareable.** The upload form requires an explicit ACL.
- **Claim inheritance.** The claim takes the most restrictive label of its evidence.
- **Server-side filtering.** The incoming view is produced by code, and the automated test asserts that leads-only and private text cannot leak.
- **Redaction preview.** The Sources page shows what an external model would see after secret and identity replacement.
- **Secret handling.** GitHub tokens are used for the request only and are never stored.
- **Model boundary.** The default extractor runs locally. If an external adapter is enabled in a later round, only team-labelled text is eligible; leads and private sources stay inside the deployment.
- **Human control.** Edits and gap answers are labelled HUMAN ASSERTED and are never presented as machine-verified.
- **No monitoring.** Baton operates on one project at the moment a handover is needed.

8. Reliability and failure handling

Failure case	What Baton does
Quote not found in the source	Marks the claim QUOTE FAILED; it stays out of the reader view and appears for correction or rejection
Two sources disagree	Marks the topic CONFLICTING and blocks publishing until a human resolves it
Task with no owner	Creates a critical MISSING gap; blocks publishing until answered or accepted
Dependency with no contact	Creates a critical MISSING gap; blocks publishing until answered or accepted
No rollback procedure	Creates a non-critical MISSING gap; shown as a known gap once accepted
No sources	Stops the pipeline with a clear message instead of producing an empty handover
File too large or unreadable	Rejects the upload with an explicit error; existing sources are untouched
GitHub import fails	Returns the HTTP status and keeps the rest of the case usable
Rebuild after human decisions	Requires confirmation and resets derived claims; the source inventory is preserved
Unapproved claims	Never included in the published handover

9. Testing evidence

The end-to-end smoke test uses a temporary database, so it never touches local data. It exercises the same routes a judge uses and asserts the permission guarantee directly.

Command: python scripts/smoke_test.py

```
SMOKE OK■
  claims: 26 (approved 25)■
  conflicts: 1■
  gaps: 3■
  labels: ['CONFLICTING', 'INFERRED', 'MISSING', 'SINGLE_SOURCE', 'VERIFIED']■
  temp data: C:\Users\hp\AppData\Local\Temp\baton-smoke-9iq4pf3x
```

What the test asserts

- The planted motor-driver conflict is detected.
- Critical gaps are detected.
- Publishing is blocked before the human decisions.
- Conflict resolution and gap handling update the publish gate.
- Publishing produces version 1.
- The incoming view does not contain leads-only or private text.
- The approver view does contain the restricted claims.
- Acknowledgement is recorded and the audit log contains the publish and acknowledge events.

10. Changes from the Round 1 plan

Round 1 plan	Round 2 implementation	Why
Next.js + FastAPI + PostgreSQL	FastAPI + Jinja + SQLite	One deployable service; the trust model and claim schema are unchanged. A React frontend can be added without changing the ledger.
Claude extraction as the primary path	Deterministic local extraction, optional LLM adapter later	The demo works without a paid key; the verifier is the source of truth, and an LLM cannot grade its own work.
Task and chat connectors in Round 3	Task CSV and chat export connectors already work	They were cheap to add and strengthen the permission demonstration.
Redis/RQ background jobs	Synchronous pipeline with visible progress	The demo corpus is small; a queue is a scale concern, not a correctness concern.
PostgreSQL row-level security	Application-level claim filtering plus an automated role test	The user-visible guarantee is tested: zero restricted claims in the incoming view.
PDF export, version diff, question flow, acknowledgement in Round 3	Acknowledgement and gap answers are partially implemented; version diff and PDF export remain Round 3	They are meaningful, visible additions for the next round.

Scope discipline. The build focused on one complete journey rather than many shallow integrations. Two source families (documents and repository exports) plus task and chat imports are enough to demonstrate the whole workflow. Live Slack or Gmail OAuth, mobile apps and enterprise SSO remain out of scope for the event.

11. Three major improvements for Round 3

1. Departing-member question flow

What: critical gaps become targeted questions routed to the departing member, with structured answers, reminders and a status for each question. Answers are stored as HUMAN ASSERTED evidence and re-verified where possible.

Why: the current build lets the approver answer or accept a gap. Routing the question to the person who actually knows closes the tacit-knowledge gap instead of forcing the approver to guess.

How measured: every critical gap has an answer or an explicit acceptance; the handover shows the answer's author and timestamp.

2. Version diff and decision timeline across publishes

What: a reader-facing 'what changed since vN' view and a diff of claims added, removed, re-labelled or re-owned between handover versions, including conflict resolutions and gap decisions.

Why: a handover is not a one-time document. When the project moves on, the incoming person needs to see what changed without re-reading the whole ledger.

How measured: republishing a case produces a visible diff with counts per section and links to the changed claims.

3. Incoming-person feedback loop and section-level acknowledgement

What: the incoming person can confirm, dispute or request access per section. A dispute re-opens the claim for the approver, and the audit log captures the full loop.

Why: acknowledgement is currently a single receipt. Section-level feedback turns the handover into a two-way, auditable transition rather than a broadcast.

How measured: a disputed section returns to the approver queue, and the audit log shows the dispute, the correction and the re-acknowledgement.

12. Demo script for judges

- Open the dashboard. The demo case shows 6 sources and 0 claims before the first build.
- Open the review console and click **Rebuild ledger**.
- Inspect the conflict card: meeting notes say driver B; README and commit say driver A.
- Click **Keep this side** on the driver-A group.
- Answer the spare-driver ownership gap with a short human answer; accept or answer the demo-venue dependency gap.
- Click **Approve all reviewable**. The publish banner turns green.
- Click **Publish handover**. The incoming view opens as approved v1.
- Note the restricted-item count on the right. The incoming person cannot see the leads-only chat or the private budget note.
- Switch to the approver role to see the restricted claims. Open the Audit page to see every decision.
- Click **Acknowledge receipt** as the incoming person; the acknowledgement appears in the audit log.

13. Known limitations and future scope

- **Authentication is simulated with role tabs** so judges can open every link without signing in. Production would bind roles to an authenticated session or SSO.
- **SQLite is single-node.** A multi-user deployment should move the ledger to PostgreSQL.
- **The deterministic extractor is strongest on structured notes and exports.** Free prose needs the optional LLM adapter or careful marker use.
- **The GitHub connector is read-only** and capped at 20 issues and 20 commits per import.

- **PDF extraction depends on a text layer.** Scanned PDFs need OCR, which is future scope.
- **The free hosting filesystem is ephemeral.** The synthetic demo reseeds after a restart; a persistent volume is needed for real data.
- **No email or notification delivery yet.** The departing-member question flow is a Round 3 improvement.
- **Future scope:** PostgreSQL and row-level security, SSO, live Slack/Drive/Jira connectors, continuous ledger mode, on-premise local models, role templates and a grounding evaluation suite.

14. Responsible design and disclosures

- All demonstration data is synthetic. No confidential workplace material is used.
- Connectors require explicit consent from the source owner; the upload screen warns against confidential material.
- Restricted claims are filtered server-side and rendered only as a count.
- Human edits and gap answers are never presented as machine-verified.
- The audit log records who approved, edited, resolved, published and acknowledged.
- The product does not monitor employees; it operates on one project at the moment a handover is needed.
- **AI assistance disclosure:** an AI assistant was used to help structure, draft and revise the plan and the implementation. The verifier, permission tests and demo behaviour are executable and can be inspected in the repository.
- **Third-party libraries:** FastAPI, Uvicorn, Jinja2, pypdf, httpx, python-docx and reportlab (report generation only). No product code existed before the build window.

15. Team and contribution

Name	Role	Contribution
Ritika Bhati	Team lead; backend, verification, frontend, product and QA	Claim schema, deterministic extractor and verifier, conflict and gap rules, FastAPI routes, SQLite ledger, permission filtering, review console, reader view, synthetic dataset, tests, documentation and demo

Team name: ByteStorm. Solo submission for CodeFrontier Hackathon, Round 2.

Appendix A. Repository structure

```
Baton/  (repository root)
  app/
    main.py          FastAPI routes and pages
    db.py            SQLite schema and helpers
    seed.py          synthetic rover-club dataset
    extract.py       typed claim extraction
    verify.py        labels, conflicts, gaps, publish gate
    pipeline.py      chunking and build orchestration
    redaction.py     secret stripping and pseudonymisation preview
    templates/       dashboard, review, sources, handover, audit
    static/          app.css, app.js
  scripts/
    smoke_test.py    end-to-end test
    make_round2_report.py
  requirements.txt
  Dockerfile
  render.yaml
  run.ps1 / run.sh
  README.md
```

Appendix B. Run and test

```
Windows:
  cd Baton
  .\run.ps1

macOS / Linux:
  cd Baton
  sh run.sh

Docker:
  docker build -t baton .
  docker run --rm -p 8000:8000 baton

Test:
  python scripts/smoke_test.py
```